

Yiannis Konstantinou

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EDUCATION

Duke University - Pratt School of Engineering, Durham, NC

Aug 2025 – May 2029

B.S.E. Electrical & Computer Engineering; Double Major in Computer Science

- **Awards:** Most Innovative Design Award, Duke Engineering Design Expo - selected among 35+ teams (Dec 2025)
- **Coursework:** Data Structures & Algorithms, Signals & Systems, Engineering Design, Computing for Engineers

TECHNICAL PROJECTS

Fish-Operated Vehicle - Computer Vision & Robotics

Sep 2026

Team of 2 - designed, built, and demoed in under 24 hours

- Designed and built a vehicle driven by a live fish; demo video reached 40,000+ views and 1,800+ likes on Instagram in 72 hours
- Implemented OpenCV pipeline estimating fish heading and swim effort in real time - dual HSV + LAB segmentation with moment-based heading estimation on Raspberry Pi 5 - translated into holonomic drive commands streamed over serial to Pico 2 W mecanum-wheel controller - tuned asymmetric slew limiting drive commands to keep water tank stable while vehicle drives

Poker Equity Calculator - Python

Aug 2026 – Present

Solo project - pure standard-library engine

- Processed 100,000+ Monte Carlo deals in under 15 seconds; validated against published equities to within $\pm 0.5\%$ across 20 benchmark matchups; wrote 15+ pytest unit tests covering evaluator edge cases (wheel straights, kicker ties, full houses)
- Built Texas Hold'em equity engine from scratch: 7-card hand evaluator with full tiebreaker logic, exhaustive enumeration for heads-up matchups, and Monte Carlo simulation for multi-opponent scenarios with partial/full board support
- Currently developing OpenCV card-recognition extension using Pillow-rendered card assets and template matching

Diabetes Alert Service-Dog System - Embedded Systems

Aug 2025 – Dec 2025

Team of 4 - dog-side hardware lead; won Most Innovative Design at Duke Engineering Design Expo

Duke University

- Achieved 91.67% end-to-end alert accuracy across 300 trials and 4.125/5 client ease-of-use rating, iterating through weekly feedback sessions with Type 1 diabetes client
- Built wireless alert system: 41 g pressure-sensing bringsel bitten by service dog triggers vibrating wrist bracelet over ESP-NOW between two XIAO ESP32-C3 boards with a battery life of over 6 hours - no phone or WiFi dependency
- Engineered dog-side hardware, integrating pressure sensor into sewn biothane housing balancing bite resistance and consistent actuation; performed most of unit's soldering and specified trigger logic (sustained-bite hold, cooldown) preventing false alerts

EXPERIENCE & LEADERSHIP

Cypriot National Guard - Lance Corporal, Mandatory Military Service

Jun 2024 – Aug 2025

14 months of active service

Nicosia, Cyprus

- Redesigned and automated camp's daily service sheet, assigning day and night patrols fairly across all soldiers - sharply cut roster-creation time and eliminated uneven duty loads, improving daily camp performance
- Promoted to Lance Corporal; instructed and evaluated ~25 recruits in weapons handling and live shooting drills under pressure, enforced unit discipline, and executed high-stakes daily camp-protection protocols
- Completed Warrior School, full-scale war simulation, and strategic planning exercises in case of hostile threat in our camp

APOEL FC Youth Team - Team Captain (Striker)

Jun 2020 – May 2024

Top national youth league squad

Nicosia, Cyprus

- Captained 11-player squad to National Championship (2022) and two National Cup titles (2022, 2024); shaped match strategy and lineup decisions with head coach and made in-game tactical calls based on experience and gut feeling
- Mentored teammates individually on positioning and mentality and helped integrate new players into the team

Red Cross Cyprus - Fundraising Volunteer, Youth Department

Aug 2023 – Jun 2024

Youth-led fundraising for disadvantaged families

Nicosia, Cyprus

- Organized fundraisers inside and outside schools for disadvantaged families - home-cooked food preparation and packaging initiatives, Christmas holiday campaigns - and hosted awareness seminars with Youth Department

Motor Oil (Hellas) — Marketing & Education Assistant intern

Jun 2022 – Jul 2022

Renewable-energy education & digital campaigns

Athens, Greece

- Executed digital campaign for Greek Refineries; led 10-person team designing renewable-energy and climate-change education programs for schoolchildren, built around interactive, game-based activities

SKILLS, ACHIEVEMENTS & INTERESTS

- **Programming:** Python, C (embedded), Java, MATLAB, Simulink
- **Tools & Hardware:** Git, OpenCV, Claude Code, Fusion 360, Raspberry Pi, ESP32 (ESP-NOW), Arduino, soldering
- **Languages:** Greek & Cypriot dialect (native), English (fluent), Spanish (basic)
- **Interests:** Football (national-level captain), chess, poker, fishing, startups & tech, padel, ping-pong,
- **A-Levels:** A* Mathematics, A* Economics, A Physics, A Geography; **IGCSEs:** four 9s (highest grade), one 8, two 7s
- **High School:** Apolytirion 99/100 (Valedictorian); honors list every year; school-highest grades in Economics and Geography